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Roll No

EC-4003-CBGS

B.E., IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Integrated Circuits and its Applications

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Define feedback topologies. Explain with diagram. 7
b) What is the concept of positive feedback and explain LC oscillator (Any one type). 7
2. a) Explain block diagram of op-amp and compare ideal and practical characteristics of op-amp. 7
b) What is differential amplifier and explain any one type in detail. 7
3. a) Explain op-amp operation as filter and describe butterworth 2nd order low pass filter. 7
b) i) Explain phase-shift oscillator. 7
ii) Derive gain of non-inverting amplifier.
4. a) Describe series clipper circuit with waveform. 7
b) Explain Astable Multivibrator. Using IC-555 timer. 7
5. a) Compare Fixed and Adjustable voltage regulator. 7
b) Write down the characteristics of standard regulator IC's in detail. 7

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6. a) Describe the amplifier with negative feedback and write its advantages. 7
b) Write down any five characteristics of op-amp in detail. 7
7. a) Explain antilog amplifier and derive the output voltage expression. 7
b) Discuss sample and hold circuit with diagram. 7
8. Write short notes on: 14
 - a) CMRR
 - b) Zero Crossing Detector
 - c) Clamper Circuit (positive peak clamper)
